

Building the DevOps Pipeline with Jenkins

Jenkins is an open source tool that provides integration with the tools used in application life cycle management to automate the entire process, based on feasibility.

The prerequisites for installing Jenkins on a specific system are:

- Jenkins official documentation recommends 256MB of RAM.
- Jenkins official documentation recommends 10GB of drive space; however, I suggest 50GB - 80GB of free space on a disk drive.
- Jenkins official documentation recommends Java 8 or 11.

The commands to run Jenkins are as follows.

You can start Jenkins at the command line by using the Generic Java package (.war):

```
java -jar jenkins.war
```

To access the Jenkins dashboard, browse to <http://localhost:8080>. Change the port to run Jenkins, with the following command:

```
java -jar jenkins.war --httpPort=9999
```

Access Jenkins' dashboard: Browse to <http://localhost:9999>

It will redirect the user to unlock the Jenkins page by providing the administrator password, which can be found from the console log when the *jenkins* command is executed for the first time.

Click on *Install Suggested Plugins*.

Once all the plugins are installed, create your first admin user (Figure 2) and click on *Save and Continue*.

Provide the Jenkins URL or keep the default, depending on your needs.

Once the Jenkins setup is complete, click on *Start using Jenkins*. Verify the Jenkins dashboard. Click on *Manage Jenkins*.

Go to *Manage Jenkins -> Global Tool Configuration*.

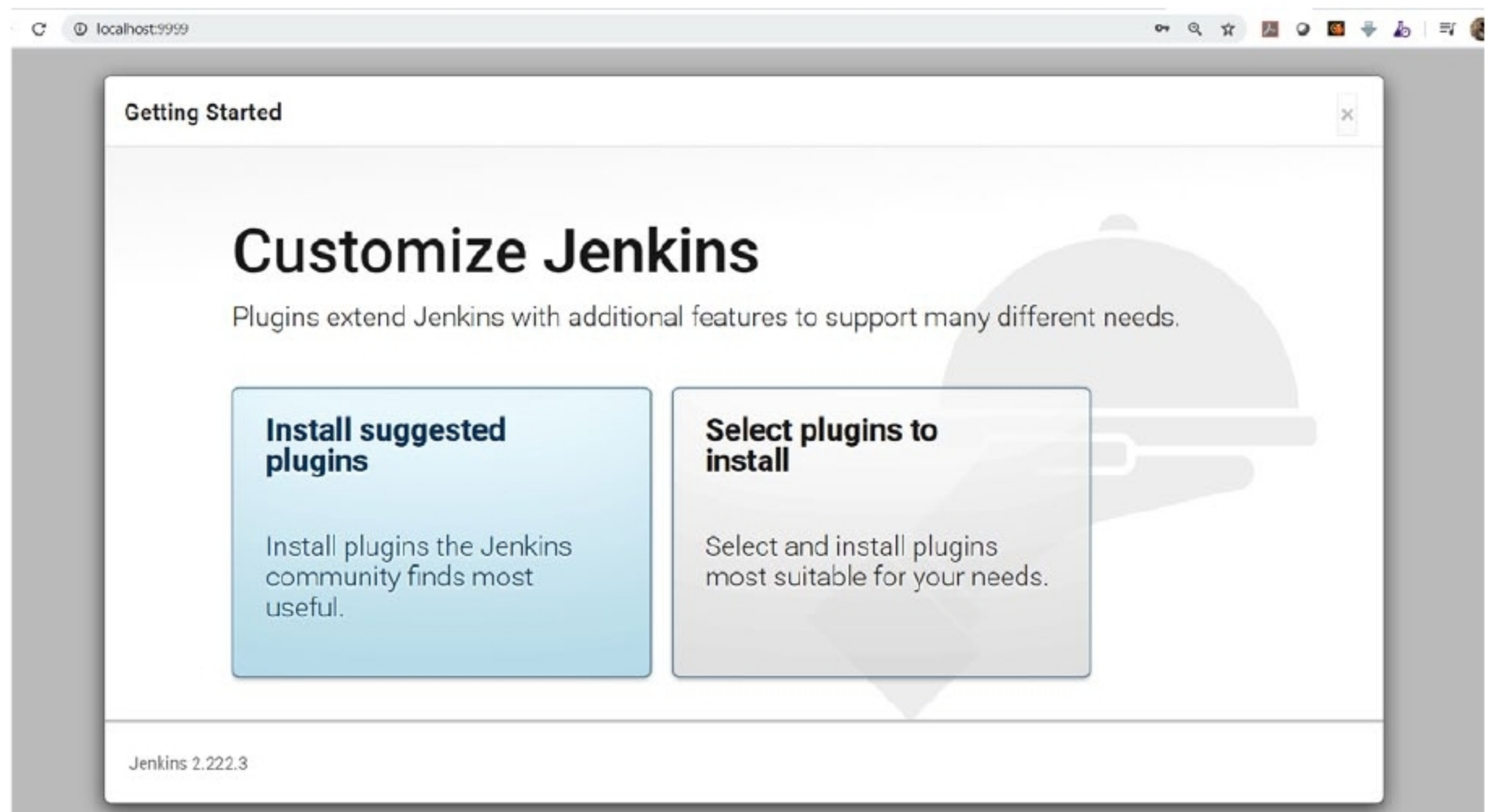


Figure 1: Customise plugins

